LESSON 6 • TERM 4 • LEVEL 1

Count Everything

A system means no misses and no repeats

TODAY WE WILL

Count with a system.

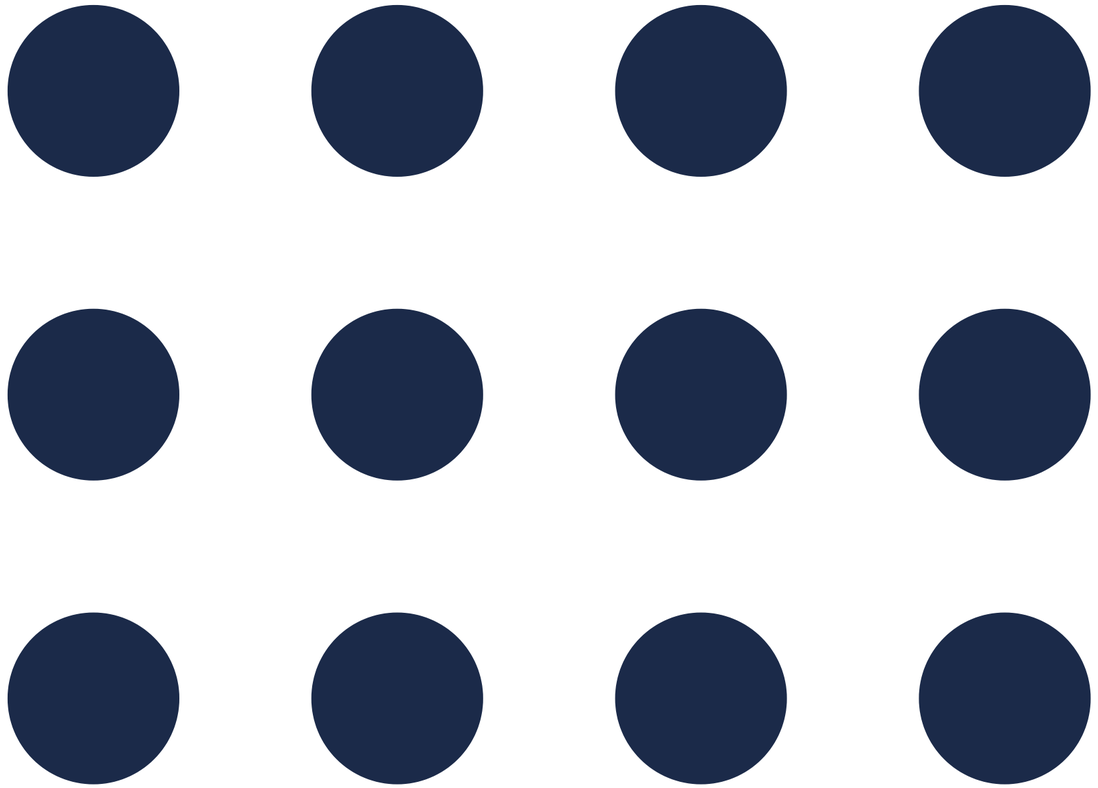
Miss none. Count none twice.

Count handshakes and squares.



Warm-up: count by rows

2–8 MIN



Count the dots by rows. How many rows? How many dots in each row? How many in all?

Warm-up: make tens

2–8 MIN

7 + 3 + 6 + 4

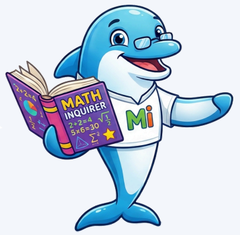
Find the pairs that make 10 first. What is the total?

2–8 MIN

Warm-up: a list with a rule

Ways to make 6 with two numbers.

Start with 1. 1 + 5 = 6. What comes next? Stop when the pairs repeat.



Take your time!

Sneak-peek puzzles

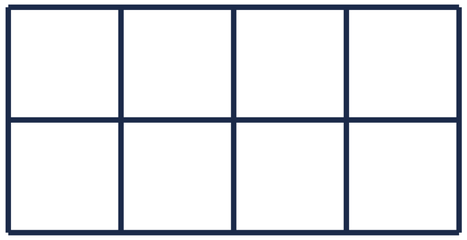
8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

How many squares are in the picture?

Write the number.



PUZZLE 2: think

Ali has 2 hats and 3 scarves.

He wears one hat and one scarf.

How many different ways can he do it?

Write or draw them.

Explore: the handshakes

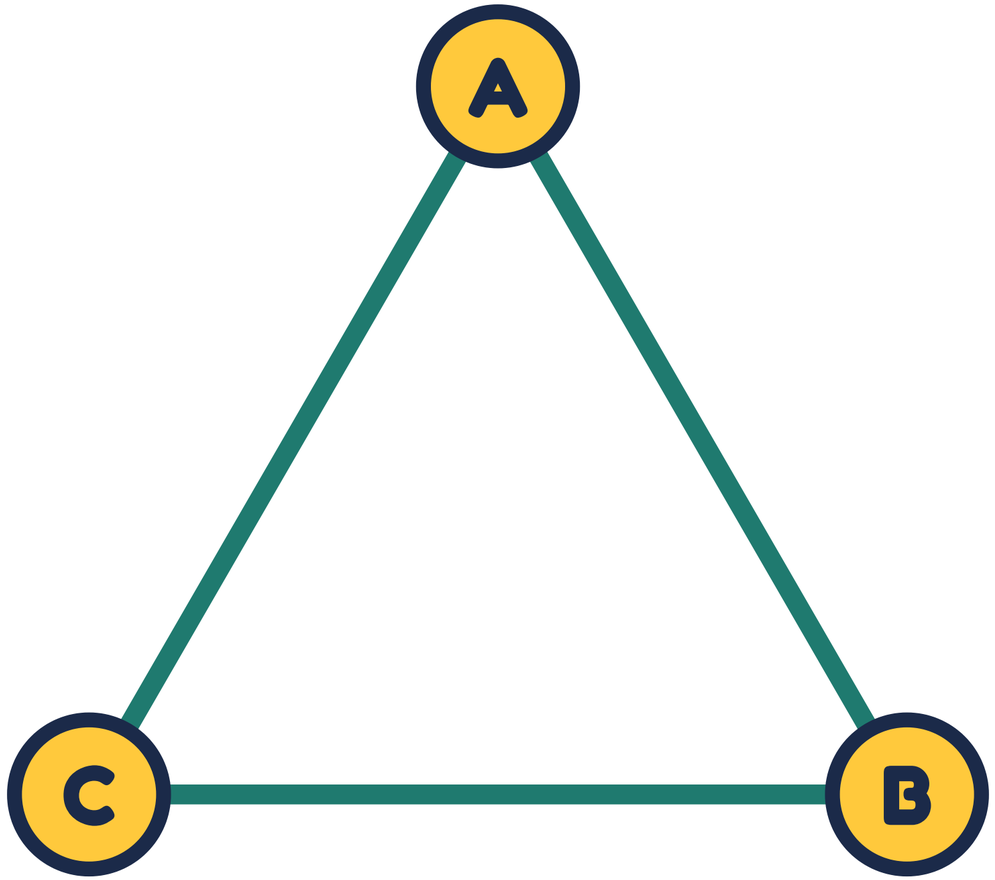
12–23 MIN

1. Stand 3 paper people up in a ring.
2. Every two people shake hands once.
3. List the handshakes. No repeats. None missed.

How many handshakes are there?

3 handshakes

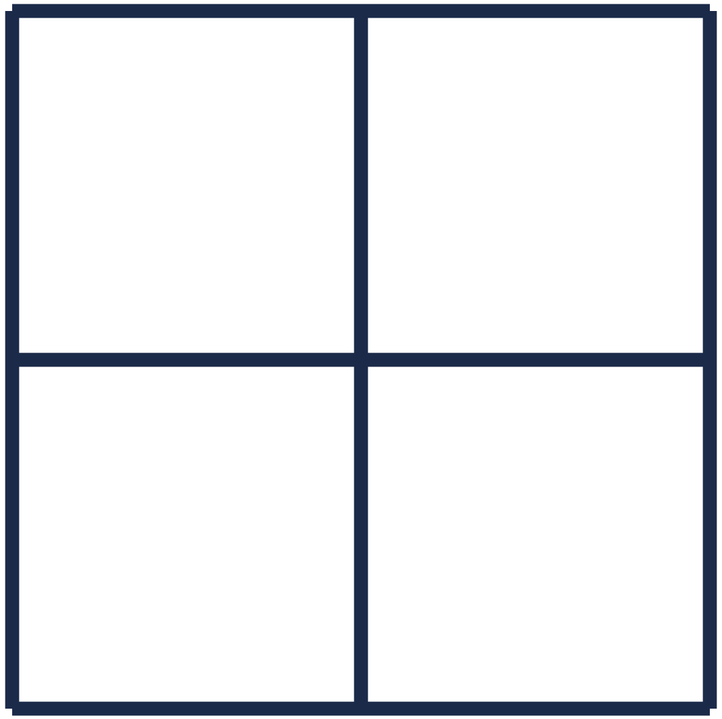
12–23 MIN



Math Inquirers Term 4 • Notice it. Test it. Explain why.

Explore: find every square

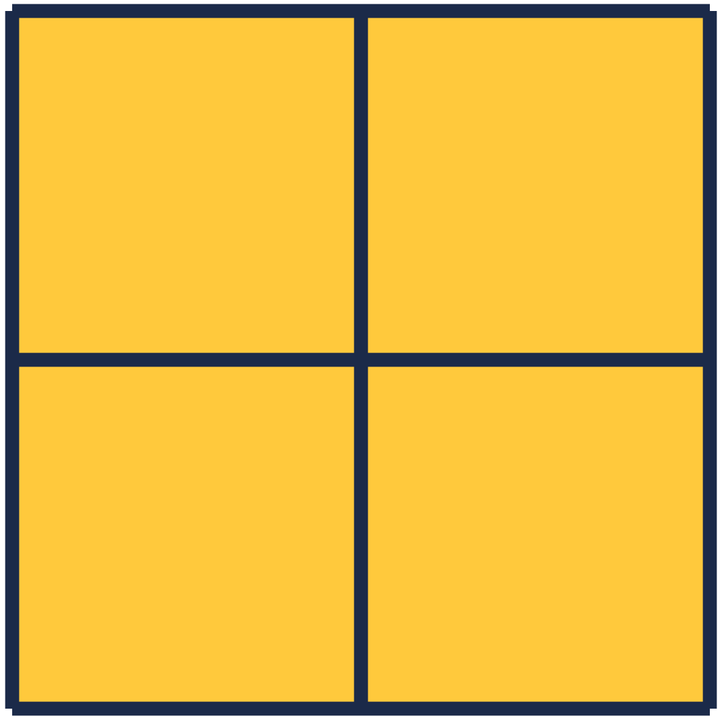
12–23 MIN



How many squares can you find? Look for small squares and a big one.

5 squares

12–23 MIN



4 small squares and 1 big square. 4 and 1 make 5 squares.

What did we notice?

12–23 MIN

- How did you know you had every handshake?
- How did you find the big square?

Count the new ones

23–34 MIN

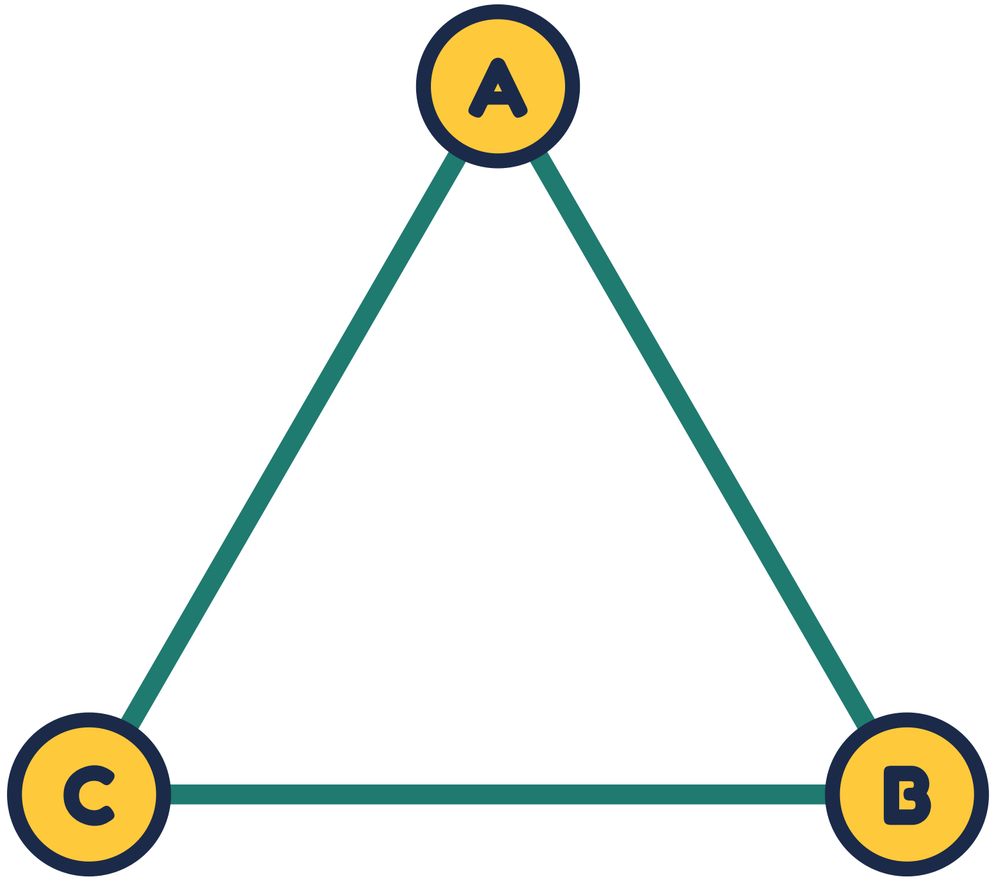






Only the new ones

23–34 MIN



A table of outfits

23–34 MIN

	skirt 1	skirt 2
☐	☐	
☐	☐	

Mei has 2 tops and 2 skirts. Each box is one outfit. How many outfits can she make?

4 outfits

23–34 MIN

	skirt 1	skirt 2
top 1	top 1 + skirt 1	top 1 + skirt 2
top 2	top 2 + skirt 1	top 2 + skirt 2

2 tops, and each top goes with 2 skirts. 2 and 2 make 4 outfits.

23–34 MIN

A list that stops

Every pair of numbers that make 8.

Start with 1. 1 + 7 = 8. What is next? When do we stop?

4 pairs

23–34 MIN

First	Second
1	7
2	6
3	5
4	4

1 + 7, 2 + 6, 3 + 5, 4 + 4. We stop at 4 + 4, because the next pairs repeat. 4 pairs.

Spot the repeat

23–34 MIN

A CHILD IN ANOTHER CLASS LISTS HANDSHAKES

A with B

A with C

B with A

B with C

She says: 4 handshakes. Is she right? What went wrong?

B with A is a repeat

23–34 MIN

B with A is the same handshake as A with B. It is counted twice. There are 3 handshakes, not 4.

34–38 MIN

Every move with every move

Stand up. Arms up or arms out.

Legs: jump or squat.

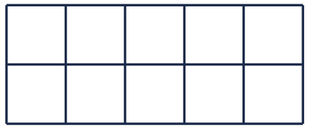
Try every arm move with every leg move.

Workbook page 1: Start here

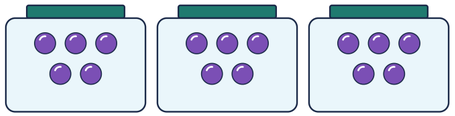
38–70 MIN

Questions 1 to 3. Take your time.

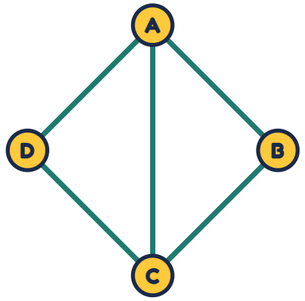
S1 Count the squares by rows.



S2 There are 3 jars. Each jar has 5 marbles.



S3 Each line joins two dots.



Check page 1

38–70 MIN

S1	10 squares
S2	15 marbles
S3	5 lines

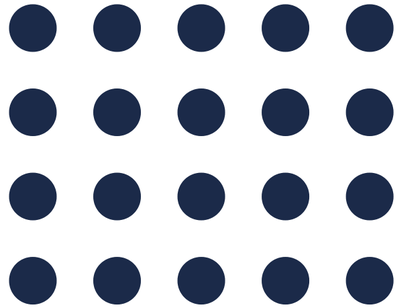
Workbook page 2: Start here

38–70 MIN

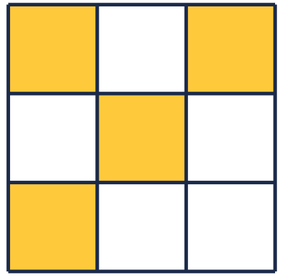
Questions 4 to 6. Take your time.

S4 Every pair of numbers makes 9. Fill in the missing numbers.

S5 Count the dots by rows.



S6 There are 9 squares. Some squares are yellow.



38–70 MIN

Check page 2

S4	7; 6; 5
S5	20 dots
S6	5 squares

38–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Handshake Party, outfits, squares of every size, pairs
Go further (page 4)	More handshakes, a bigger grid, two roads
Investigation (page 5)	Every pair that makes 7
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

Four friends shake hands.

Each one shakes hands with 3 others.

So there are 4 times 3 handshakes.

Do you agree? How could we check?

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: 8 squares. Puzzle 2: 6 ways. Each of the 2 hats goes with each of the 3 scarves.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Count the squares by rows.**

There are ☐ squares.



E2 **Each box is one way to wear a hat and a scarf.**

There are ☐ ways.

	scarf 1	scarf 2	scarf 3
hat 1	☐	☐	☐
hat 2	☐	☐	☐
hat 3	☐	☐	☐

E3 **Four friends, A, B, C and D, make pairs. Here are 5 of the 6 pairs: AB, AC, AD, BC, BD.**

The missing pair is _____.

Check the exit ticket

76–82 MIN

E1	15 squares
E2	9 ways
E3	C and D

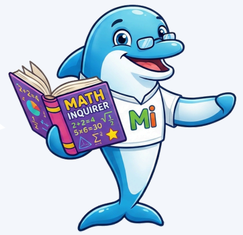
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ handshakes ☐ squares ☐ the list of pairs ☐ something else
What helped me	☐ my paper people ☐ a system ☐ a list ☐ I was not stuck
Next time I will try	☐ counting with a system ☐ handshakes ☐ listing pairs ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

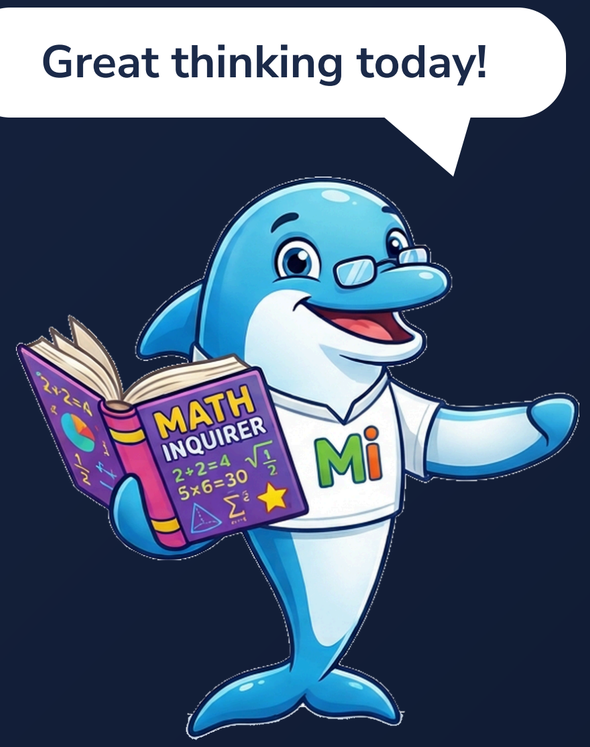
Well done, inquirers!

Ask a grown-up: how many handshakes at a party of 4?

NEXT LESSON

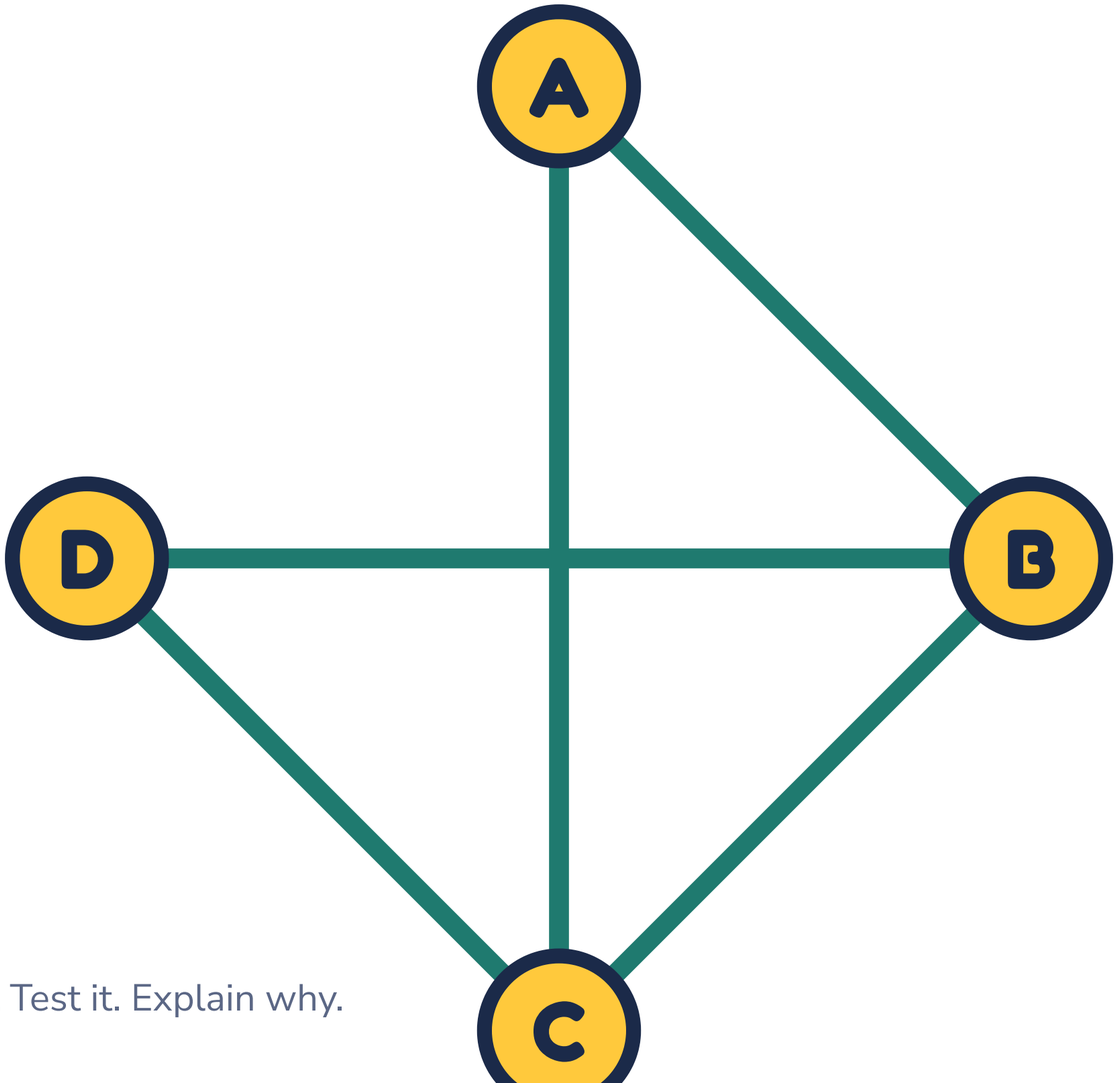
Make Your Own Graph

Have ready: Book 8 and about 12 small things from the house to sort.



Go further: which handshake is missing?

OPTIONAL



OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Does a bigger group always make more handshakes?
Make a puzzle	Make a counting puzzle for your grown-up.
Teach it back	Tell a toy how to count every handshake once.

Optional. Everyone may join.